Worksheet 6–2 B Riemann Sums #2
AP Calculus Name_____

t (hours)	0	1	3	5	7
$R(t)$ (people/hour)	4250	4000	3125	2000	750

A local amusement park measured the rate $R(t)$, measured in people per hour, that people were entering the park at various times t, where t is measured in hours since 8AM. $R(t)$ is decreasing for $0 \leq t \leq 7$ hours.

1. Use a right Riemann Sum with the four subintervals indicated in the table above to approximate $\int_0^7 R(t)dt$. Include units of measure.

2. Is your answer in Question 1 an over or under approximation of $\int_0^7 R(t)dt$? Give a reason for your answer.

3. Using correct units, interpret the meaning of $\int_0^7 R(t)dt$ in context of the problem.

4. Use a trapezoidal approximation with four subintervals to approximate $\int_0^7 R(t)dt$.

t (sec)	0	2	4	7	10	11	12
$v(t)$ (m/s)	4	0	-3	4	2	6	5

For $0 \leq t \leq 12$ seconds, a bug is walking along a horizontal axis with velocity $v(t)$, measured in meters per second. Selected values of the differentiable function $v(t)$ are shown in the table above.

5. Use a midpoint Riemann sum with the intervals $[0, 4]$, $[4, 10]$, and $[10, 12]$ to approximate $\int_0^{12} v(t)dt$. Using correct units, interpret the meaning of $\int_0^{12} v(t)dt$ in the context of the problem.

6. Use a left Riemann sum with the four subintervals indicated in the table to estimate $\int_2^{11} v(t)dt$.

7. Explain why there must be a time c, $0 < c < 7$, when the acceleration of the bug is zero.

8. The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by the twice-differentiable and strictly increasing function $R(t)$, where t is time in minutes. A table of selected values of $R(t)$ for the time interval $0 \leq t \leq 90$ minutes is shown below. At $t = 0$ the plane had already consumed 84 gallons of fuel.

Time (minutes)	0	30	40	50	70	90
$R(t)$ (gallons per min)	20	30	40	55	65	70

(a) Use data from the table to find an approximation for $R'(45)$. Show the computations that led to your answer. Indicate units of measure.

(b) Using a trapezoidal approximation with five subintervals, approximate how much fuel the plane has consumed after 90 minutes.

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9. Approximate the area under the curve of $f(x) = \frac{6}{x} + 5$ on the interval $[1, 3]$ using a trapezoidal approximation with $n = 3$ subintervals.

Directions: Given the scenarios below, using correct units, interpret the following in context of each situation.

On one particular morning, the number of students that enter a school is changing at a rate of $E(t)$ where t is the number of minutes since 8AM.

Two friends are playing in a game of tag. The rate that the distance between the two friends is changing can be modeled by the function $f(t)$, where $f(t)$ is measured in feet/second.

At the end of a high school basketball game, the number of fans in the stands is changing at the rate $A'(t)$, where t is the number of seconds since the game ended.

10. $\int_{2}^{11} f(t)dt = 1.2$

11. $\int_{1}^{7} A'(t)dt = -32$

12. $\int_{0}^{8} E(t)dt = 527$